How Rich Are Mixture Attitudes?

Theory and Experimental Evidence

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Oxford

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Mixtures in economics

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Consumption
bundles of goods

Risk / uncertainty
distributions over outcomes

Information
Blackwell structures

Time
consumption streams

Social
allocations

Strategy
mixed strategies

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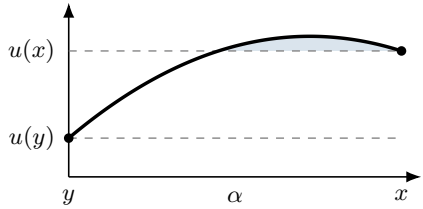
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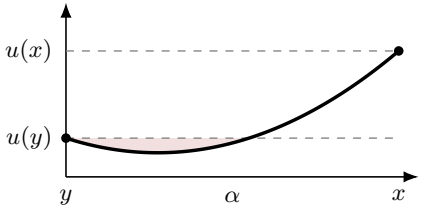
Strategy
mixed strategies

Most economic environments combine objects through a *mixture operation*

Mixture operation describes preferences

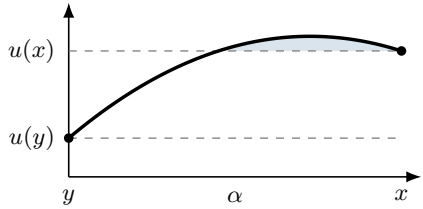


Mixture loving

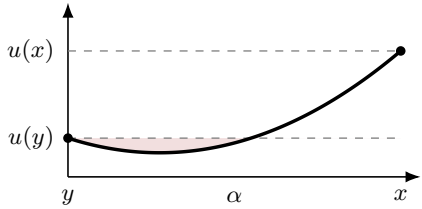


Mixture averse

Mixture operation describes preferences



Mixture loving



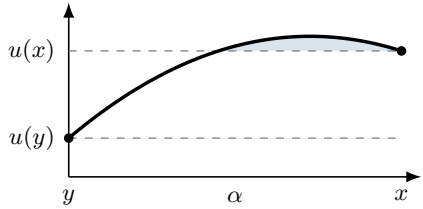
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Most theories assume one-sided attitudes

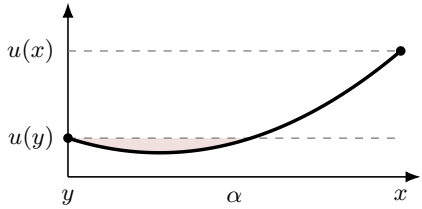
Cobb–Douglas, CES
diversification, smoothing
fairness, maxmin EU

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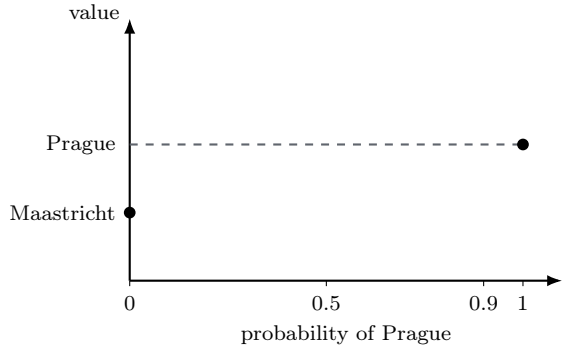
specialization
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This paper: what about preferences that are neither?

Our interest: richer mixture attitudes

Example: surprise trips

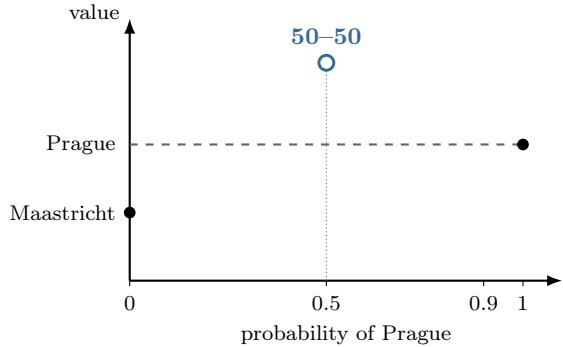
- Prague $\succ$ Maastricht



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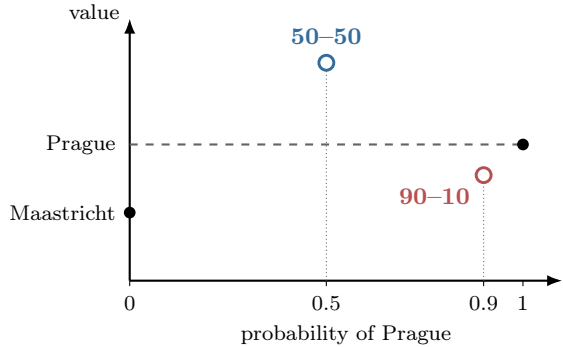
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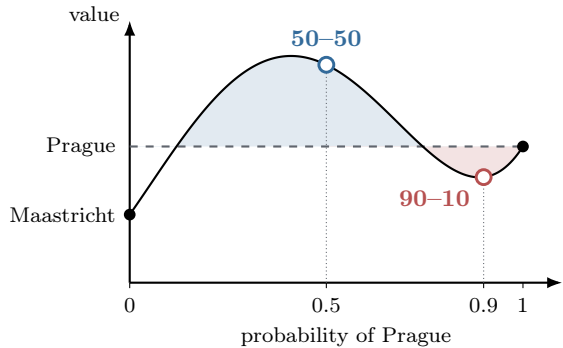
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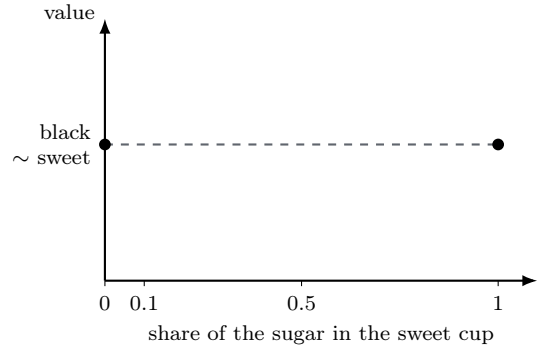


Like to randomize if sufficiently random, else prefer certainty

More examples: sugar in coffee

Consumption

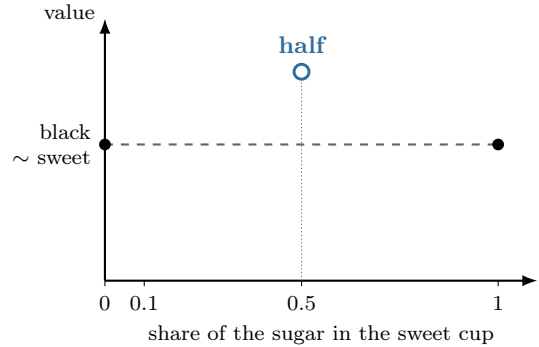
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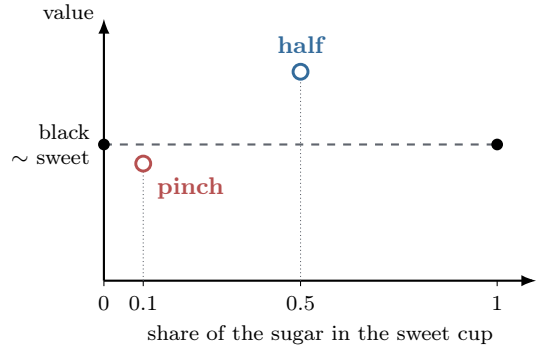
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- Half the sugar $\succ$ both



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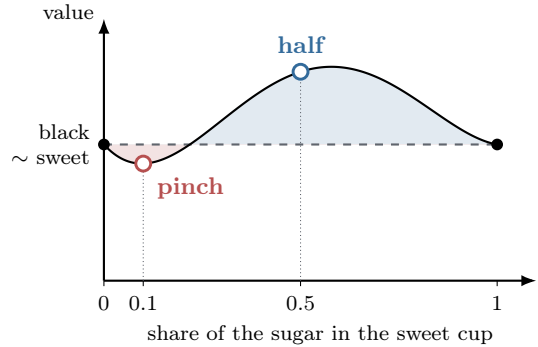
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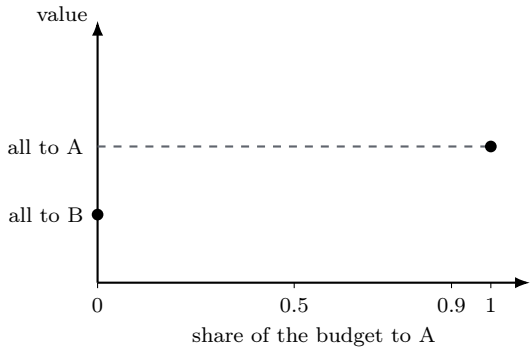


Taste needs enough sugar to register; health pays for every grain

More examples: giving to charity

Social preferences

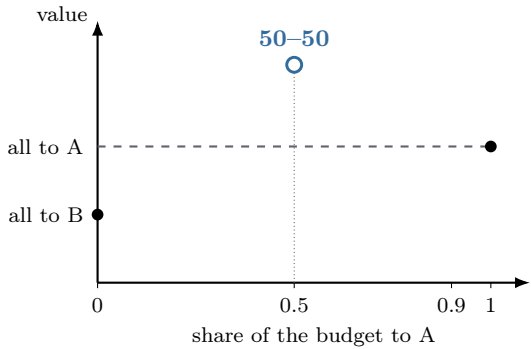
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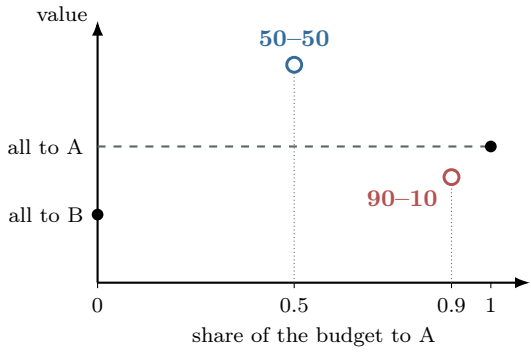
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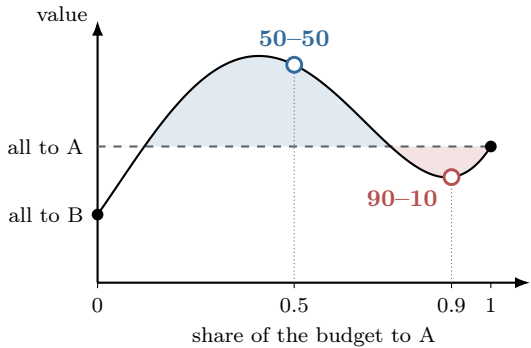
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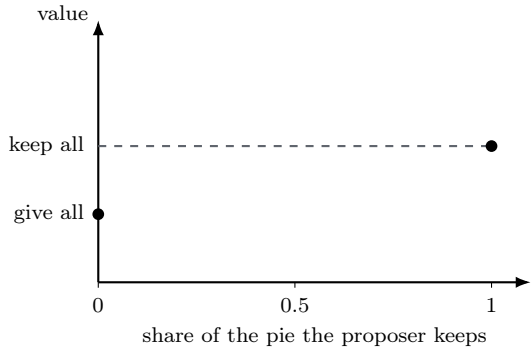
Trade-off between fairness and effectiveness: a small gift to B helps neither

More examples: the ultimatum game

Strategy

Proposer offers a share of the pie.
Responder accepts or rejects.

- Keep all $\succ$ give all
  keeping all is rarely accepted

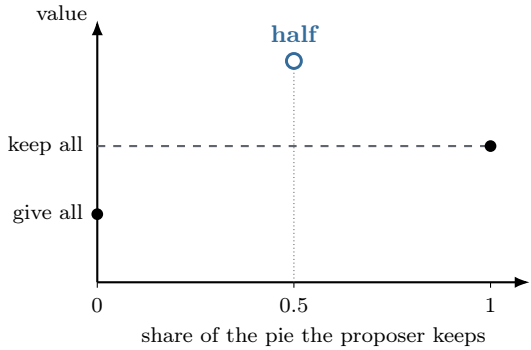


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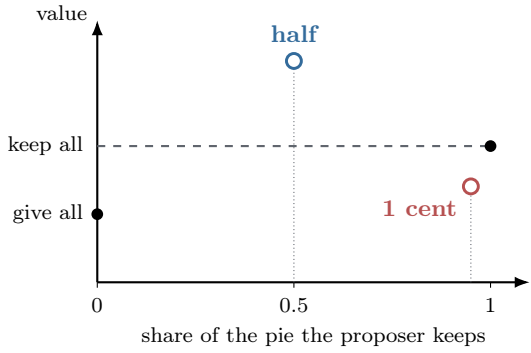


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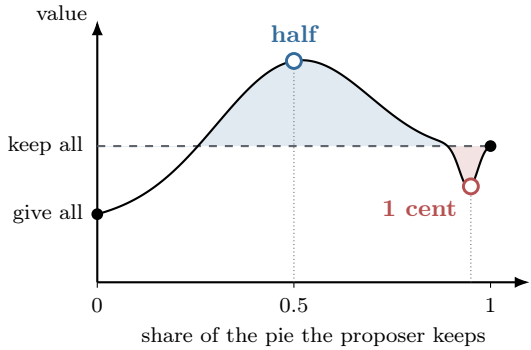


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A fair offer buys acceptance, but a token offer is an insult

Why richer mixture attitudes?

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Theory: No conceptual/organizing framework for mixed attitudes

We build a hierarchy to study and classify mixed attitudes, revealing new insights

Empirics: No evidence or tests of relevance/prevalence of mixed attitudes

Show mixed attitudes are common and missed in a previously well-studied setting

Related literature and contribution

1. Mixture and convex spaces

Herstein & Milnor (1953); Debreu (1959); Gilboa & Schmeidler (1989); Fehr & Schmidt (1999)

We propose a hierarchy to understand mixed attitudes in any mixture space

Our literature audit: very few experimental designs can detect this behavior

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New results on known models, and a way to compare their mixture attitudes

3. Preferences for randomization and for certainty

Allais (1953); Loewenstein (1987); Agranov & Ortoleva (2017); Cerreia-Vioglio et al. (2019)

Both together: a strong departure from one-sidedness, from anticipation and trade-offs across attributes

Theory

Framework

- **Objects:** compact convex $\mathcal{X} \subseteq [0,1]^d$ containing the diagonal $(t, \ldots, t)$
- **Mixing operation:** $x \alpha y = \alpha x + (1 - \alpha) y$

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Environment	**Object** x	**Mixture** $x \alpha y$	**Diagonal** $(t, \ldots, t)$
Bundles	quantities of d goods	average each good	t of every good
Allocations	payoffs of d people	average each payoff	equal split
Lotteries over $o_1 \succ o_2 \succ o_3$	$x = (p_1,\ p_1 + p_2)$ e.g. $(.5, .2, .3) \mapsto (.5, .7)$	mix the probabilities	o_1 with prob. t, else o_3

Preferences: Basic axioms

Preferences $\succsim$ on the mixture space $\mathcal{X}$:

- **Complete:** $x \succsim y$ or $y \succsim x$, for all x, y
- **Transitive:** $x \succsim y$ and $y \succsim z \Rightarrow x \succsim z$
- **Mixture continuous:** $\{\alpha : x\alpha y \succsim z\}$ and $\{\alpha : z \succsim x\alpha y\}$ are closed
- **Strictly monotone:** $x \geq y \Rightarrow x \succsim y$, and $x > y \Rightarrow x \succ y$

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Irrational, incomplete, or stochastic choice can imply strange mixture attitudes.
Our focus: intentional, rational choice.

Basic mixture attitudes

How have we thought about mixing so far? **One-sided.**

Definition (Convex: mixture loving)

For all $x \succsim y$ and $\alpha \in [0,1]$,

$$x \alpha y \succsim y.$$

Definition (Concave: mixture averse)

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This is strong! We want a hierarchy that subsumes it and expands it.

A hierarchy of mixture attitudes: MMA

Definition (Mixed mixture attitudes, MMA)

For some $x^+ \succsim y^+$, $x^- \precsim y^-$ and $\alpha, \beta \in (0,1)$,

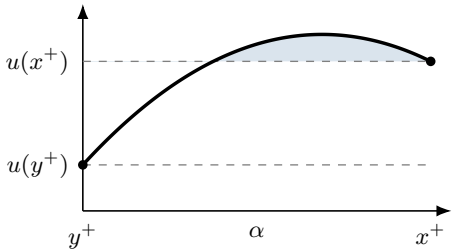
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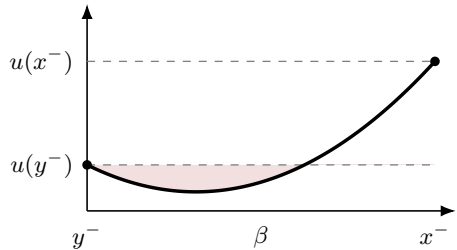
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A hierarchy of mixture attitudes: WNB

Definition (Weak non-betweenness, WNB)

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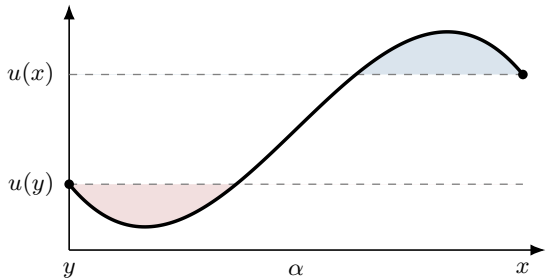
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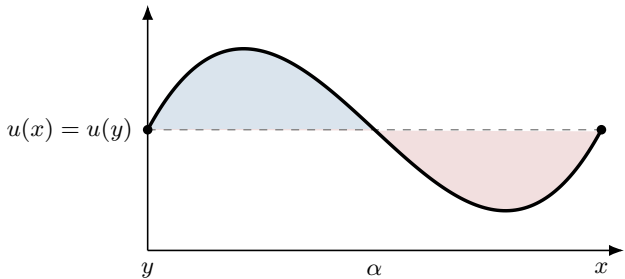
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How big is each set?

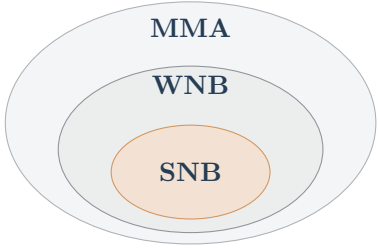
Taking stock of the hierarchy

MMA $x^+\alpha y^+ \succ x^+$ and $x^-\beta y^- \prec y^-$,

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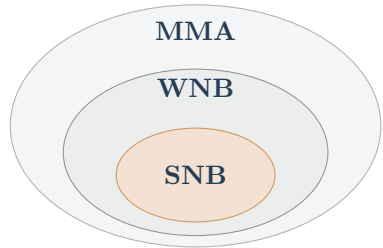
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Two-sided attitudes: it does.

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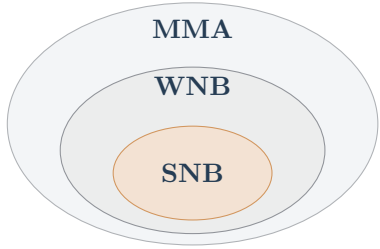
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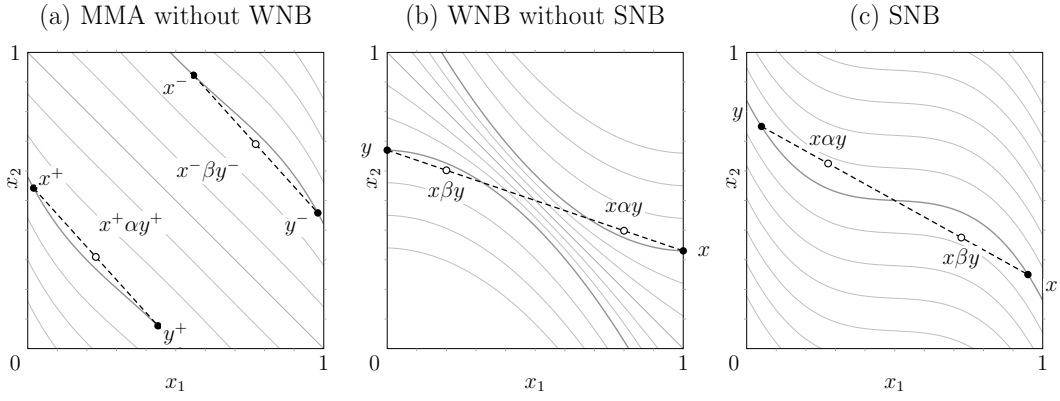
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What is in MMA without WNB? In WNB without SNB? Are these sets big?

Three example indifference maps

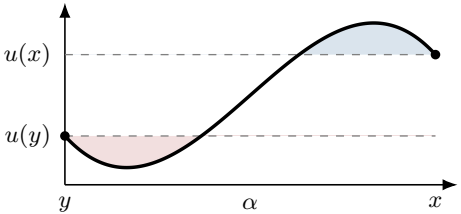


MMA $\supsetneq$ WNB $\supsetneq$ SNB.

WNB vs. SNB: the order of peak and dip

Proposition (Ordered attitudes, informal)

Without SNB, a chord $x \succ y$ with both attitudes must dip near y before it peaks near x.

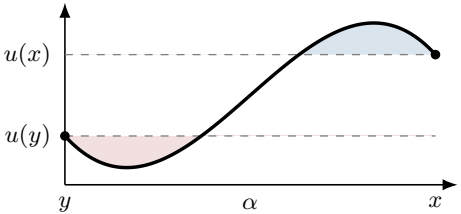


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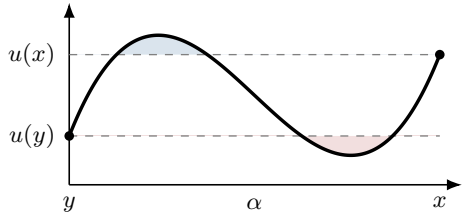
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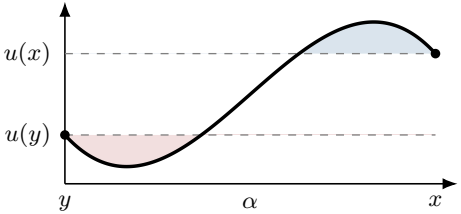


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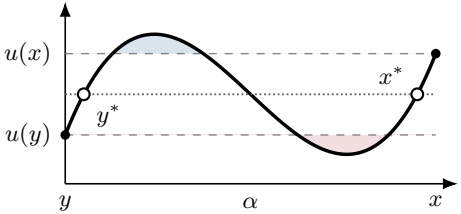
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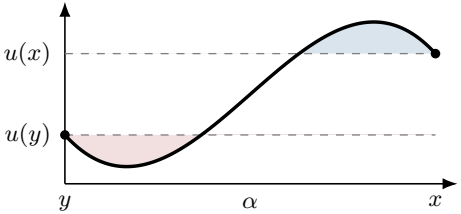


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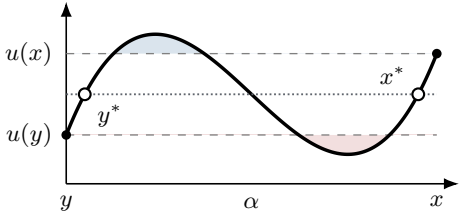
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Mixed attitudes without SNB are restrictive.
Is SNB too permissive / pathological / knife-edge?

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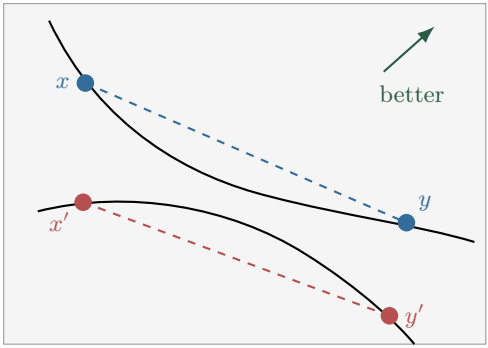
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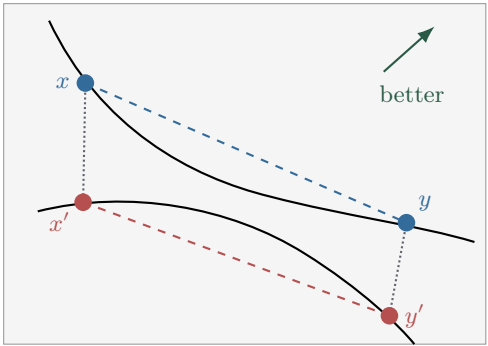
$\mathcal{P}^S$ is open and dense in $\mathcal{P}^M$.

Why SNB is generic: proof sketch



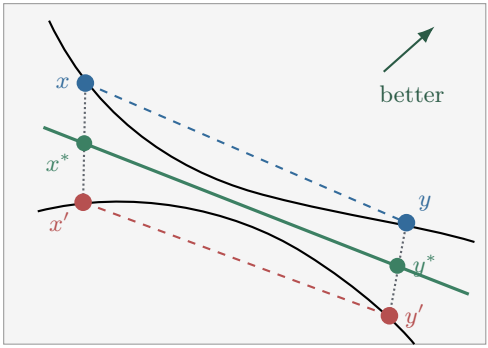
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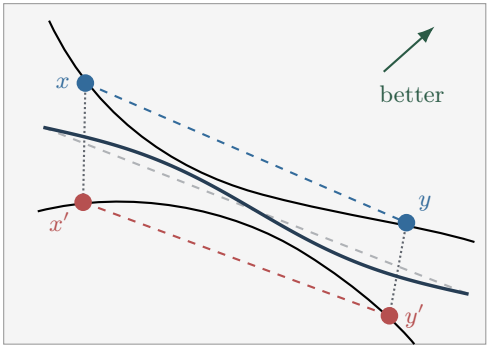
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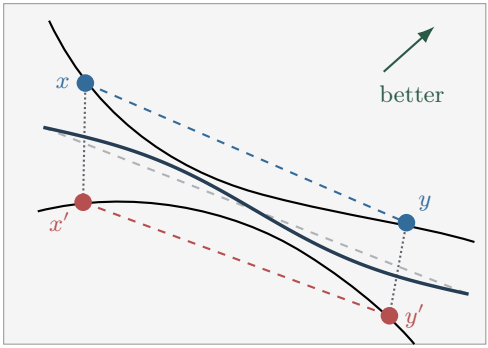
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SNB comes from distorting a neutral chord.

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Exact neutrality is the knife edge. But what if a model builds it in?

When neutrality is built in

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Certainty independence: every constant act is a neutral benchmark.
α-maxmin, Choquet and invariant biseparable: MMA $\iff$ SNB.

Classifying models

Uncertainty	MMA, not WNB	WNB, not SNB	SNB
α-maxmin EU	–	–	✓
Choquet EU	–	–	✓
Invariant biseparable	–	–	✓
Smooth ambiguity	✓	✓	✓
Variational, no UA	✓	✓	✓

Certainty-independent models display mixed attitudes only as SNB.

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Risk	MMA, not WNB	WNB, not SNB	SNB
Quadratic utility	✓	–	–
Mixture symmetry	✓	✓	–
RDEU, $n \geq 4$	–	–	✓
Pure risk	✓	–	✓
CPT	✓	✓	✓

- Masatlioglu & Raymond (2016): Quadratic utility $\cap$ RDEU$\neq \emptyset$
- We show: only if mixture attitudes are globally one-sided

▸ Quadratic utility

Why does mixed mixture attitude arise?

- Theory is explicitly *general* and *descriptive* – *why* may be domain dependent
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- Theory is explicitly *general* and *descriptive* – *why* may be domain dependent
- Experiment focuses on a specific environment to discuss particular channels
- General takeaway: rich mixture attitudes arise out of rich tradeoffs/considerations

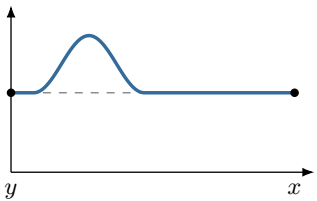
Rich motives generate SNB

Two one-sided motives: loving $\succsim_\ell$ and averse $\succsim_a$
Aggregate: $U = F(U_\ell, U_a)$, with F continuous, increasing and $F(t,t) = t$

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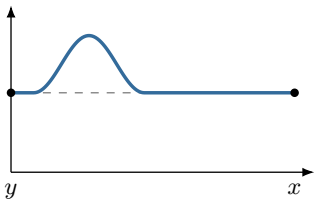
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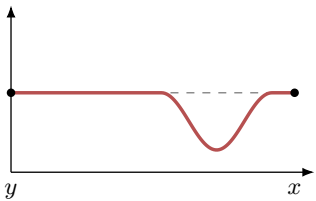
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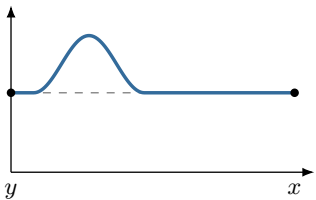
Averse motive



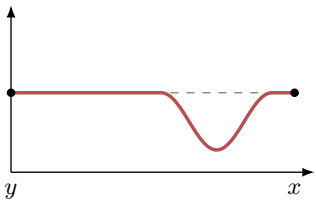
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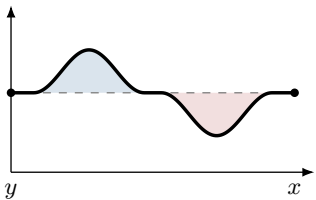
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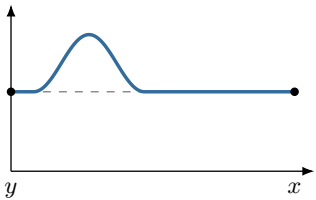


peak, then dip: SNB

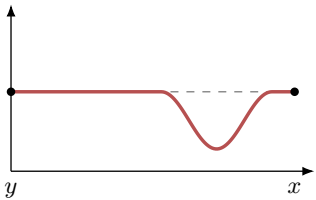
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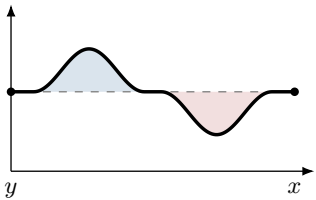
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peak, then dip: SNB

Proposition

There exist $\succsim_\ell$ and $\succsim_a$ such that the aggregate displays SNB for every such F.

Experiment

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Existing designs **cannot identify** mixed attitudes. What do we need?

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3. For WNB and SNB: several mixtures on one chord

Most designs observe a single interior mixture

Audit of experimental literature

Domain	N	**Weak Loving/Averse**	**Strict Loving**	**Strict Averse**	**MMA**	**WNB/SNB**
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Consumption budgets	23	**15** (65%)	**12** (52%)	**4** (17%)	**4** (17%)	**4** (17%)
Time preferences	85	**74** (87%)	**8** (9%)	**3** (4%)	**3** (4%)	0
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Where data allow it, SNB appears: 6 of 9 subjects in Hey & Strazzera (1989).

Experimental environment

We revisit Toussaert (2024)

Subjects choose between trip destinations and lotteries over trip destinations

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Empirical relevance

A real product: surprise-trip planning was a USD 1.4 billion market in 2024

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5. How does multidimensionality affect mixing?

Structure of the study

83 participants, London, March 2020. Online survey, 44 minutes on average

Part 1 **Trip preferences**

- Rank the 10 trips
- Value each trip, £0–500

Part 1: ten trips

Destination: **Turku** (Finland)





Description: Step back through the ages in this medieval town, and explore the maze-like chambers of the impressive Turku Castle. Enjoy nature in national parks, sunbathe on beaches, and explore museums, art galleries, historic sites, restaurants aplenty, and shopping hotspots, all around town.

Accommodation: Arrive at Helsinki airport and take a train to Turku. For the best experience of living local, stay in a cosy Airbnb that you'll have entirely to yourself, located conveniently in the central of town.

Ten European destinations

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All worth £420: cheaper travel means better accommodation.

▸ Valuations

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Strict ranking confirmed for 84% of adjacent pairs.

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Part 2 **45 binary choices**

- 6: sure trip vs. sure trip
- 29: sure trip vs. lottery, $p \in \{.1, .5, .9\}$
- 10: larger lotteries, held out

Choices: sure trips vs. surprise lotteries

After ranking the trips, $x_1 \succ x_2 \succ \cdots \succ x_{10}$, face binary choices, including:

- x_1 vs. $p\,x_1 + (1-p)\,y$ and y vs. $p\,x_1 + (1-p)\,y$
- $p \in \{0.1,\, 0.5,\, 0.9\}$
- $y \in \{x_2,\, x_3,\, x_5,\, x_{10}\}$
- Surprise: the destination is revealed in the week of departure

Choices: sure trips vs. surprise lotteries

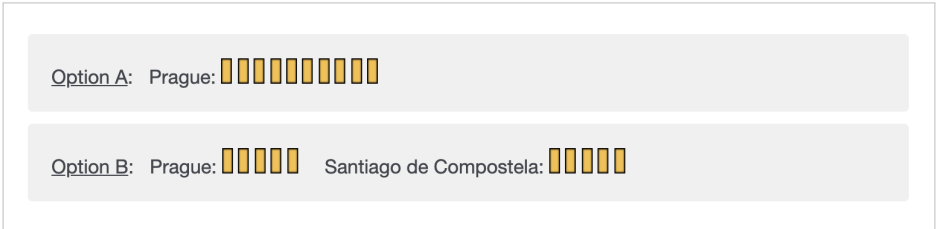
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Each mixture is observed at three weights, against both endpoints.

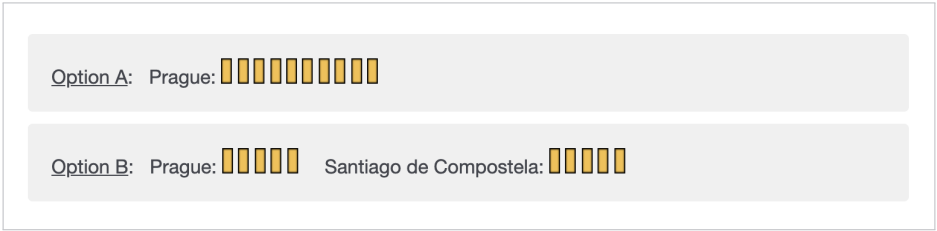
▸ All 45 problems

Part 2: what a choice looks like



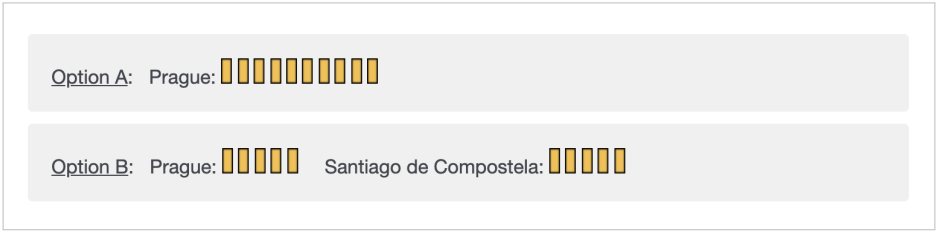
- Each ticket is a 10% chance

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- Option B: the destination comes in a sealed envelope, about 7 days before departure

Part 2: what a choice looks like



- Each ticket is a 10% chance
- Option B: the destination comes in a sealed envelope, about 7 days before departure
- Suggested: open it at the airport

Structure of the study

83 participants, London, March 2020. Online survey, 44 minutes on average

Part 1 **Trip preferences**

- Rank the 10 trips
- Value each trip, £0–500

Part 2 **45 binary choices**

- 6: sure trip vs. sure trip
- 29: sure trip vs. lottery, $p \in \{.1, .5, .9\}$
- 10: larger lotteries, held out

Part 3 **Demand for surprise**

- Build a lottery, pick the reveal date
- Value a wildcard trip, rate motives

Part 4 **Money and tastes**

- Same choices over money
- Risk, ambiguity, trip attributes

One decision is paid: 1 in 20 wins a trip. Everyone receives a £25 voucher.

Testing SNB

Definition (Strong non-betweenness, SNB)

For some $x \sim y$ and $\alpha, \beta \in (0,1)$,

$$x\alpha y \succ x \sim y \succ x\beta y.$$

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Definition (SNB*)

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Definition (SNB*)

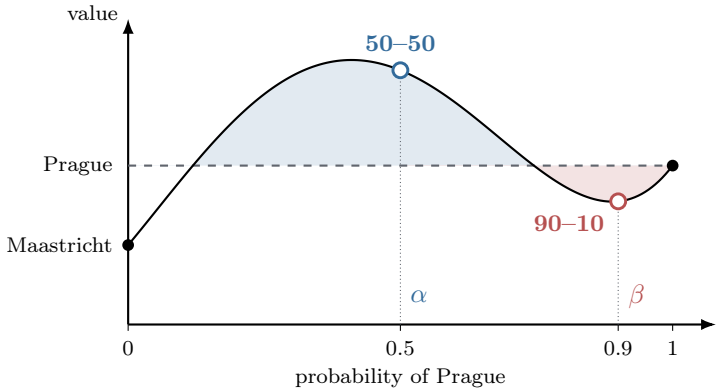
For some $x \succsim y$ and $\alpha < \beta$ in $(0,1)$,

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Proposition

Under the basic axioms, SNB $\iff$ *SNB**.

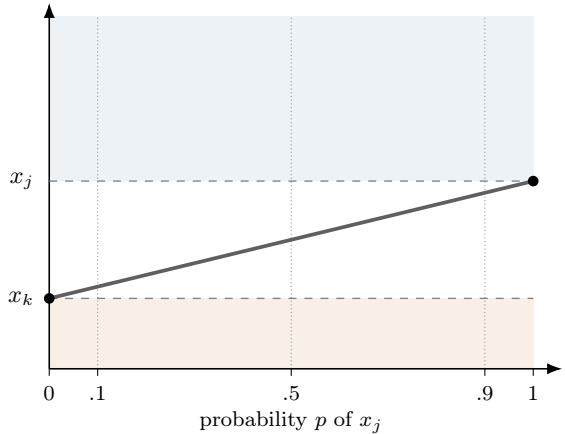
Graphical illustration



$$\alpha < \beta \;\Rightarrow\; \text{SNB}^* \iff \text{SNB}$$

From choices to types

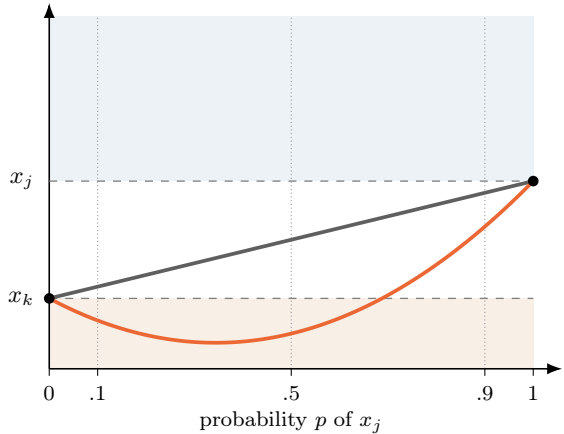
On a chord $x_j \succ x_k$, with $L(p) = (x_j, p;\, x_k, 1-p)$:



Betweenness
no lottery outside $[x_k, x_j]$

From choices to types

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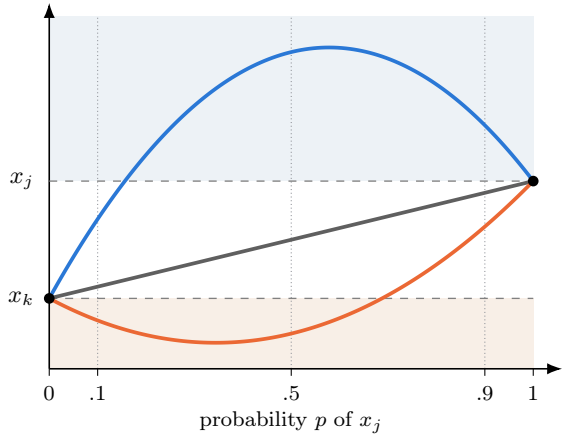


Betweenness
no lottery outside $[x_k, x_j]$

Averse only
$x_k \succ L(p)$ for some p

From choices to types

On a chord $x_j \succ x_k$, with $L(p) = (x_j, p;\, x_k, 1-p)$:



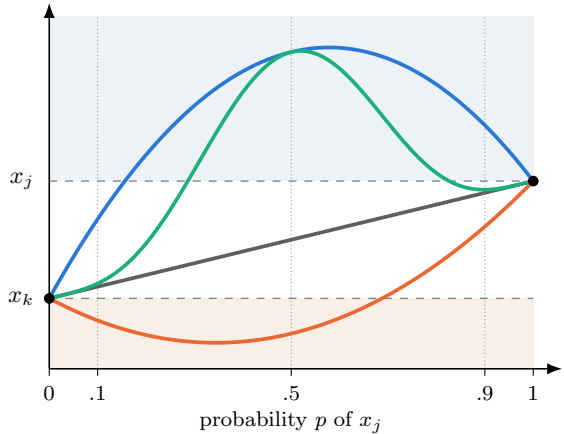
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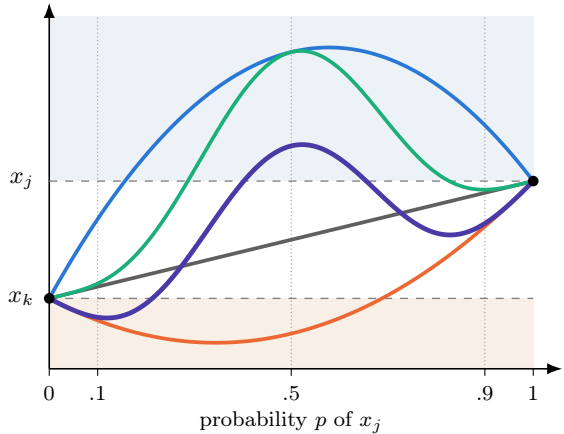
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SNB, no dip
$L(p) \succ x_j \succ L(p')$, $p < p'$
e.g. $L(.5) \succ x_j \succ L(.9)$

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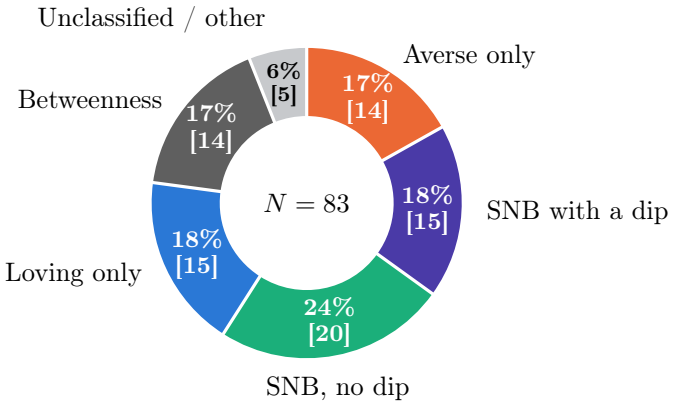
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SNB with a dip
SNB, and also $x_k \succ L(p)$

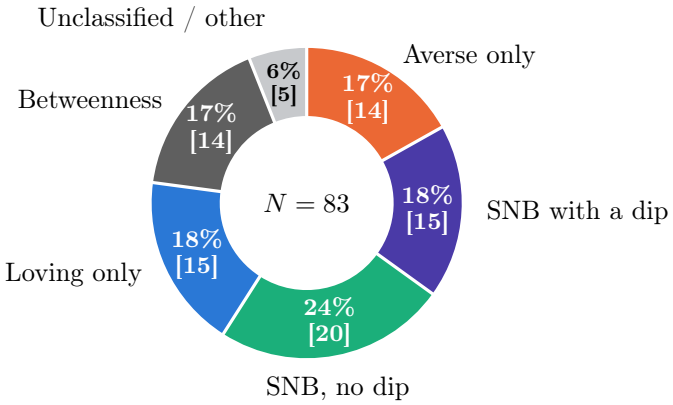
RQ1: How rich are mixture attitudes?

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▸ Strictness

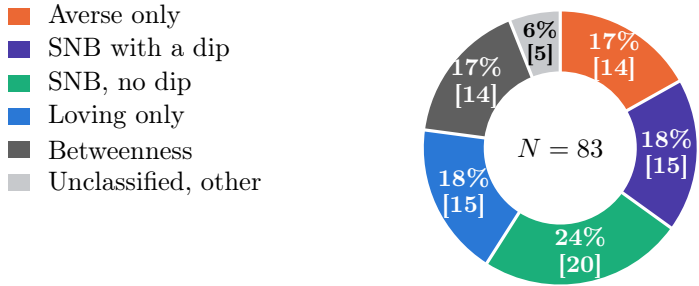
RQ1: How rich are mixture attitudes?



42% display SNB. The rest split evenly: betweenness, averse only, loving only.

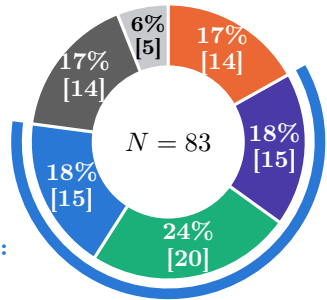
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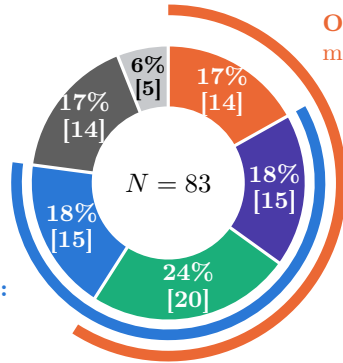
- Averse only
- SNB with a dip
- SNB, no dip
- Loving only
- Betweenness
- Unclassified, other



Only 50–50 observed:
misidentified as loving

RQ1: How rich are mixture attitudes?

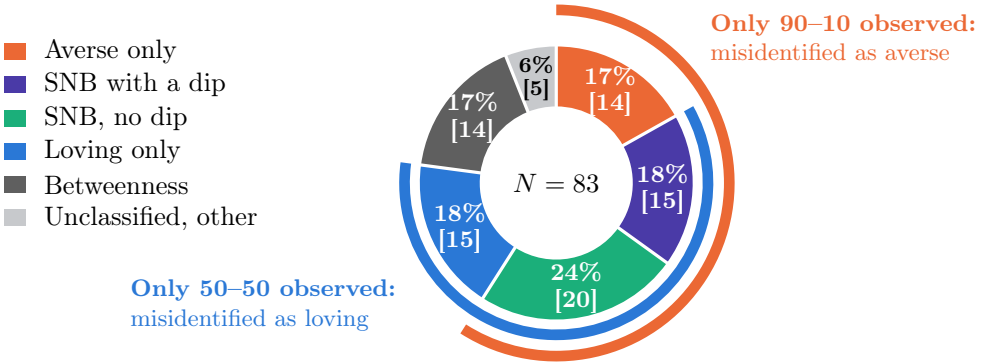
- Averse only
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Only 90–10 observed:
misidentified as averse

Only 50–50 observed:
misidentified as loving

RQ1: How rich are mixture attitudes?



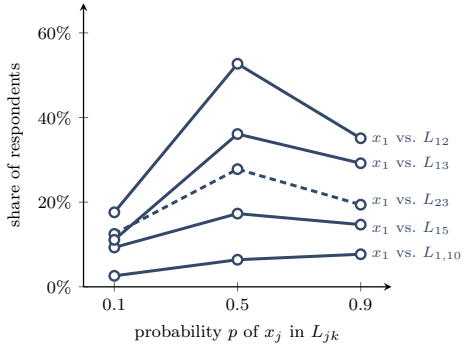
Observing a single mixture misreads SNB as a one-sided attitude.

RQ2: Which mixtures do people prefer?

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Share of participants; $L_{jk} = (x_j, p;\, x_k, 1-p)$

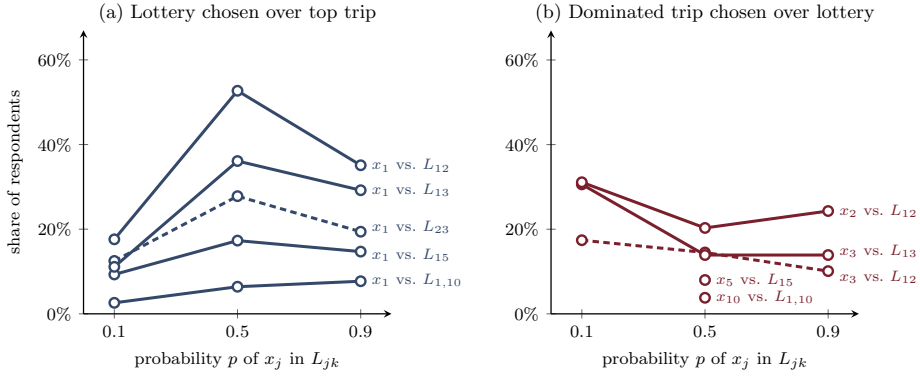
(a) Lottery chosen over top trip



▸ A mistake? ▸ Regressions

RQ2: Which mixtures do people prefer?

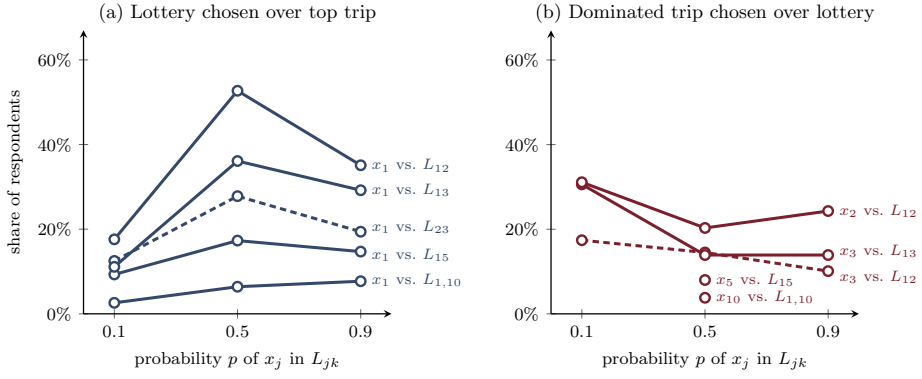
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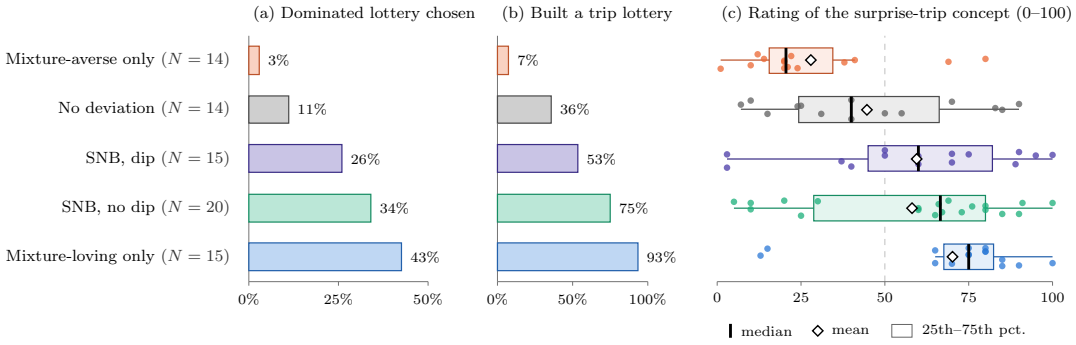


Mixing helps most at 50–50, and hurts most at 10–90.

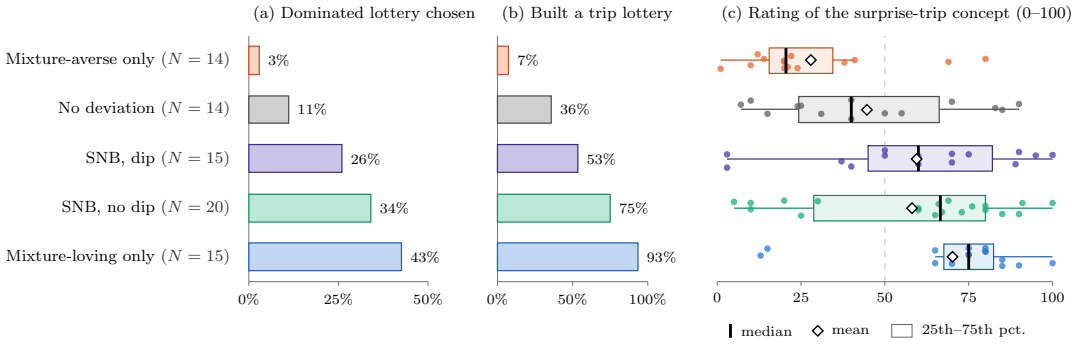
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RQ3: Is mixing behavior intentional or noisy?

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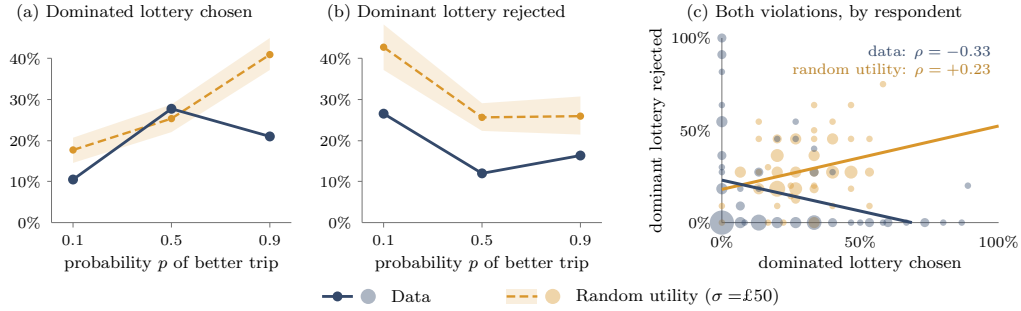
RQ3: Is mixing behavior intentional or noisy?



Types predict behavior in other choices.

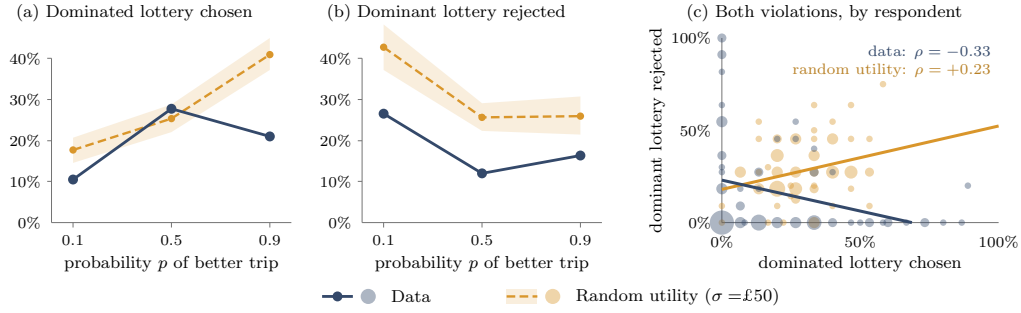
RQ3: Is mixing behavior intentional or noisy?

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▸ Benchmarks

RQ3: Is mixing behavior intentional or noisy?



Behavior cannot be rationalized by noise.

▸ Benchmarks

RQ4: Where does the (dis)taste for randomization come from?

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Share of participants ranking each reason first

What people like

rate the concept 50 or more ($N = 48$)

Excitement until the reveal	**33%**
The unique holiday experience	**27%**
Daydreaming about destinations	**19%**

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Stress of going unprepared	**26%**
Inability to plan activities	**17%**

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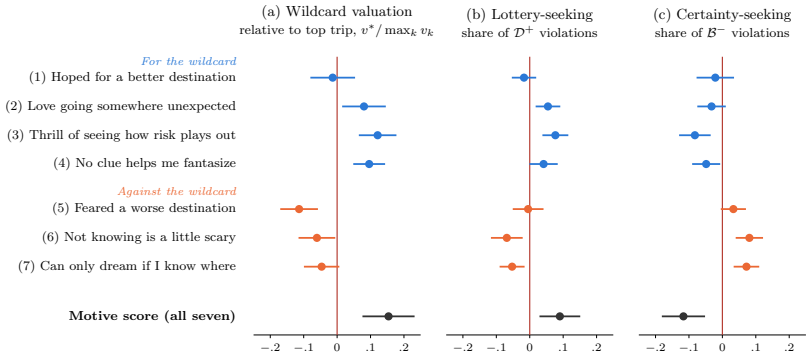
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Two forces: open questions are fun, but hard to prepare for.

RQ4: Where does the (dis)taste for randomization come from?

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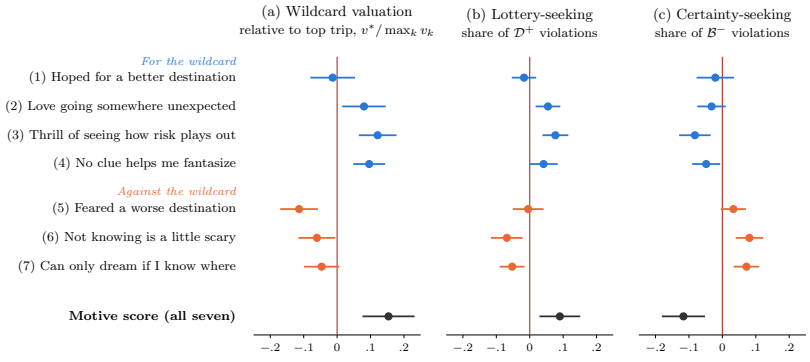
Value a **wildcard** trip to an unknown destination; rate motives for and against



▸ Lottery building

RQ4: Where does the (dis)taste for randomization come from?

Value a **wildcard** trip to an unknown destination; rate motives for and against



Motives for/against uncertainty predict wildcard value and lottery choice.

▸ Lottery building

RQ4: Where does the (dis)taste for randomization come from?

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Part 3: build your own lottery, then choose when to learn the destination

46 of 83

build a lottery

Lottery building

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74%

of builders delay the reveal, by 94 days on average

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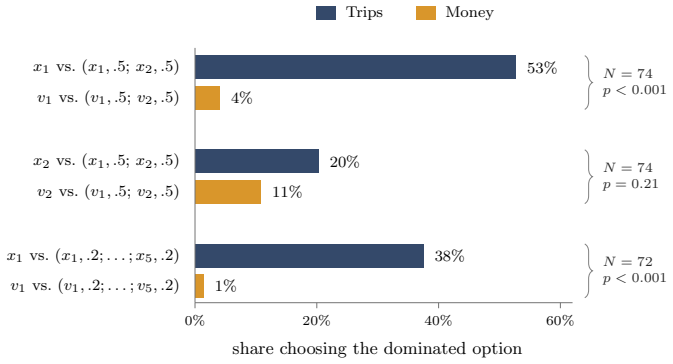
People pay to keep the surprise, but their lotteries favor top trips and drop the worst.

▸ Lottery building

RQ5: How does multidimensionality affect mixing?

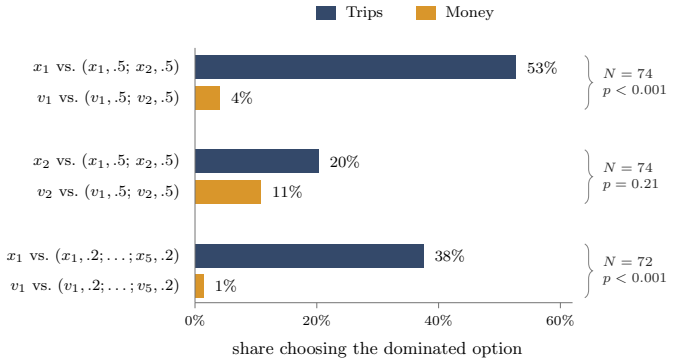
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Same choices, with each trip x_k replaced by its own valuation v_k



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Same choices, with each trip x_k replaced by its own valuation v_k



Money is one-dimensional: mixing disappears, 53% → 4%.

RQ5: How does multidimensionality affect mixing?

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Why? Trips are bundles of attributes

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RQ5: How does multidimensionality affect mixing?

Why? Trips are bundles of attributes

- Money is unidimensional: **no mixing**
- Trips have multiple attributes: we measure **9 bipolar attributes**
  - e.g. warm vs. cold, vibrant vs. quiet, urban vs. rural, beach vs. mountains

RQ5: How does multidimensionality affect mixing?

Why? Trips are bundles of attributes

- Money is unidimensional: **no mixing**
- Trips have multiple attributes: we measure **9 bipolar attributes**
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- We measure each subject's preference on each attribute:
  - **Vertical**: a clear preference for one pole
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Why? Trips are bundles of attributes

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  - e.g. warm vs. cold, vibrant vs. quiet, urban vs. rural, beach vs. mountains
- We measure each subject's preference on each attribute:
  - **Vertical**: a clear preference for one pole
  - **Horizontal**: open to either experience
- Vertical attributes rank only 9% of trip pairs;
  90% of subjects have a horizontal attribute

RQ5: How does multidimensionality affect mixing?

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A simple model: what a lottery L over two trips is worth

$$V(L) = \underbrace{\mathbb{E}_L[v]}_{\text{value}} + \underbrace{S(h)}_{\text{fun of not knowing}} - \underbrace{C(h)}_{\text{cost of not preparing}}$$

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- v: **value** of a trip, from its vertical attributes

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- Two forces: $S(h)$, the fun of not knowing; $C(h)$, the cost of not preparing

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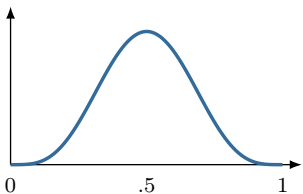
h: uncertainty about horizontal attributes v: value over vertical attributes

RQ5: How does multidimensionality affect mixing?

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Surprise S



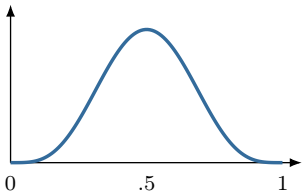
needs enough uncertainty

RQ5: How does multidimensionality affect mixing?

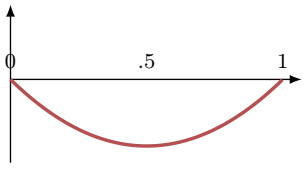
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h: uncertainty about horizontal attributes v: value over vertical attributes

Surprise S **Preparation** $-C$



needs enough uncertainty



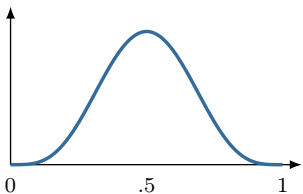
bites as soon as anything is open

RQ5: How does multidimensionality affect mixing?

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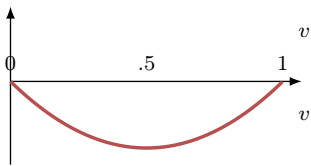
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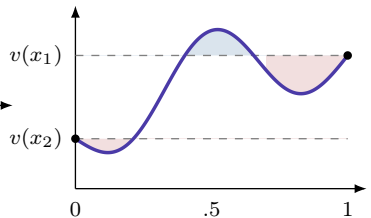
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bites as soon as anything is open

Value of the lottery

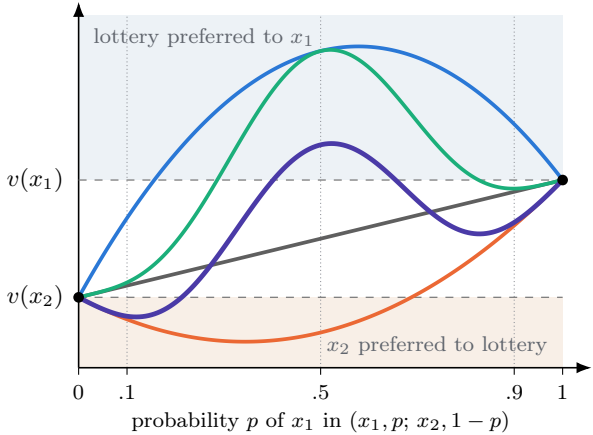


peak at 50–50, dips near certainty

RQ5: How does multidimensionality affect mixing?

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The model generates the five patterns



Betweenness
destination only

Averse only
preparation cost

Loving only
linear surprise

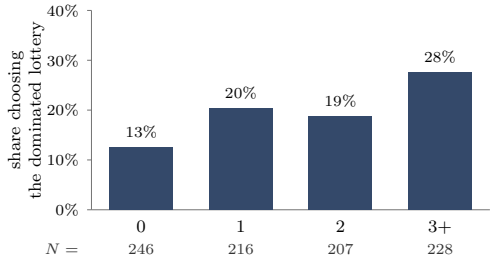
SNB, no dip
convex surprise

SNB with a dip
convex surprise + preparation

RQ5: How does multidimensionality affect mixing?

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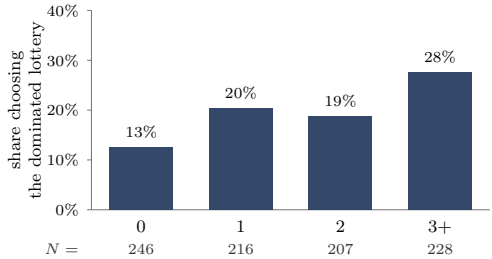
Horizontal attributes: mixing



horizontal attributes on which the two trips differ

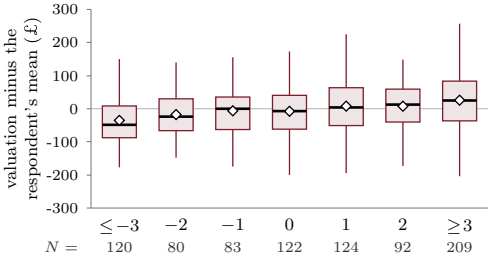
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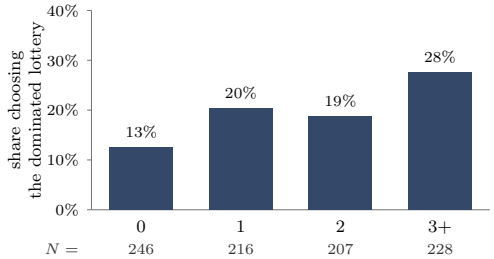
Vertical attributes: value



taste fit of the trip on vertical attributes

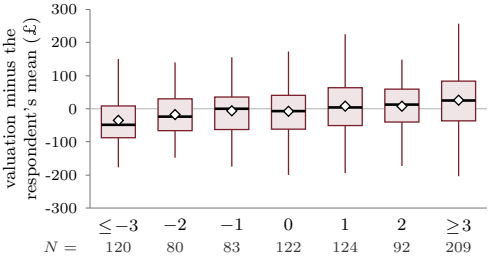
RQ5: How does multidimensionality affect mixing?

Horizontal attributes: mixing



horizontal attributes on which the two trips differ

Vertical attributes: value



taste fit of the trip on vertical attributes

Horizontal differences drive mixing: +3.8 points each.
Vertical fit drives value: +£8.4 per point.

What drives SNB? Rich/Multiplicity of Motives

Our reading: a taste for randomization when randomness is high, traded off against a preference for certainty near the 100-0.

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Theory

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Experiment

Suggestive evidence from elicited motives for choice and various other choices

Conclusion

Mixture attitudes are likely richer than we thought:

1. **Theoretically**, rich taxonomy, not knife-edge
2. **Experimentally**, common but understudied

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Many other domains/models of study left

Appendix

Why quadratic utility stops at MMA

Quadratic utility: $V(\mathbf{p}) = \mathbf{p}' \, \Phi \, \mathbf{p}$. Along any chord,

$$V(\mathbf{p}\alpha\mathbf{q}) = \alpha V(\mathbf{p}) + (1-\alpha)V(\mathbf{q}) + K(\mathbf{p},\mathbf{q})\,\alpha(1-\alpha), \qquad K(\mathbf{p},\mathbf{q}) = -(\mathbf{p}-\mathbf{q})' \, \Phi \, (\mathbf{p}-\mathbf{q}).$$

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Attitudes depend on *which* lotteries are mixed, not on *how much*.
So 50–50 surprise $\succ$ Prague $\succ$ 90–10 surprise is ruled out.

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A mistake?

Read alone, 50–50 $\succ$ Prague violates first-order stochastic dominance in the participant's own ranking (**?**). But these choices are:

Costly	$\approx$**10%** of a trip's value given up to keep the surprise
Systematic	**53%** choose the surprise at 50–50, but only **35%** at 90–10
Valued	**54%** of lottery builders pay $\geq$ £5 to delay the reveal
Specific to trips	**4%** choose the 50–50 lottery when prizes are money

Trips are bundles. All cost £420, so a cheaper city means a better hotel. Without attribute-wise dominance, monotonicity is silent. Money is the one-attribute limit.

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A choice from our experiment

Sure trip

your favorite destination

(say, Prague)

vs.

50–50 surprise
Prague or your 2nd favorite
(say, Maastricht)

53%
choose the surprise

90–10 surprise
90% Prague, 10% Maastricht

35%
choose the surprise

Many make both choices: 50–50 surprise $\succ$ Prague $\succ$ 90–10 surprise

How strict is the SNB classification?

Choices reveal weak preferences. We read choices at the design points as strict.

SNB participants	**35**
Strict by transitivity through strictly ranked trips	9
Overturned only by two exact ties	8
Overturned by one exact tie	18

Most conservative reading: 9 of 83 participants (11%) display SNB.

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The 45 decision problems

DP #	Core problems (35 choices)	**Use**
	Degenerate lotteries (6 choices)	
1, 4, 10, 13	A: $(x_1, 1)$ vs. B: $(x_k, 1)$ for $k \in \{2, 3, 5, 10\}$	Ranking
7	A: $(x_2, 1)$ vs. B: $(x_3, 1)$	Ranking
16	A: $(x_5, 1)$ vs. B: $(x_{10}, 1)$	Ranking
	Binary $\mathcal{D}^+$ problems (15 choices)	
2, 18, 27	A: $(x_1, 1)$ vs. B: $(x_1, p; x_2, 1-p)$ for $p \in \{0.5, 0.9, 0.1\}$	ML
5, 20, 29	A: $(x_1, 1)$ vs. B: $(x_1, p; x_3, 1-p)$ for $p \in \{0.5, 0.9, 0.1\}$	ML
8, 22, 31	A: $(x_1, 1)$ vs. B: $(x_2, p; x_3, 1-p)$ for $p \in \{0.5, 0.9, 0.1\}$	ML*
11, 24, 33	A: $(x_1, 1)$ vs. B: $(x_1, p; x_5, 1-p)$ for $p \in \{0.5, 0.9, 0.1\}$	ML
14, 25, 34	A: $(x_1, 1)$ vs. B: $(x_1, p; x_{10}, 1-p)$ for $p \in \{0.5, 0.9, 0.1\}$	ML
	$\mathcal{B}^-$ problems (11 choices)	
3, 19, 28	A: $(x_2, 1)$ vs. B: $(x_1, p; x_2, 1-p)$ for $p \in \{0.5, 0.9, 0.1\}$	MA
6, 21, 30	A: $(x_3, 1)$ vs. B: $(x_1, p; x_3, 1-p)$ for $p \in \{0.5, 0.9, 0.1\}$	MA
9, 23, 32	A: $(x_3, 1)$ vs. B: $(x_1, p; x_2, 1-p)$ for $p \in \{0.5, 0.9, 0.1\}$	MA*
12	A: $(x_5, 1)$ vs. B: $(x_1, p; x_5, 1-p)$ for $p = 0.5$	MA
15	A: $(x_{10}, 1)$ vs. B: $(x_1, p; x_{10}, 1-p)$ for $p = 0.5$	MA
	Sure trip between lottery outcomes (3 choices)	
17, 26, 35	A: $(x_5, 1)$ vs. B: $(x_1, p; x_{10}, 1-p)$ for $p \in \{0.5, 0.9, 0.1\}$	Level
	Out-of-sample problems (10 choices)	
	$\mathcal{D}^+$ over 3, 5 or 10 trips (9 choices)	
36	A: $(x_1, 1)$ vs. B: $(x_1, 0.4; x_2, 0.3; x_3, 0.3)$	ML
37	A: $(x_1, 1)$ vs. B: $(x_1, 0.4; x_3, 0.3; x_5, 0.3)$	ML
38	A: $(x_1, 1)$ vs. B: $(x_2, 0.4; x_3, 0.3; x_4, 0.3)$	ML*
39	A: $(x_5, 1)$ vs. B: $(x_6, 0.4; x_7, 0.3; x_8, 0.3)$	ML*
40	A: $(x_1, 1)$ vs. B: $(x_1, 0.2; x_2, 0.2; x_3, 0.2; x_4, 0.2; x_5, 0.2)$	ML
41	A: $(x_1, 1)$ vs. B: $(x_1, 0.2; x_3, 0.2; x_5, 0.2; x_7, 0.2; x_{10}, 0.2)$	ML
42	A: $(x_1, 1)$ vs. B: $(x_6, 0.2; x_7, 0.2; x_8, 0.2; x_9, 0.2; x_{10}, 0.2)$	ML*
43	A: $(x_5, 1)$ vs. B: $(x_6, 0.2; x_7, 0.2; x_8, 0.2; x_9, 0.2; x_{10}, 0.2)$	ML*
44	A: $(x_1, 1)$ vs. B: $(x_1, 0.1; x_2, 0.1; ...; x_9, 0.1; x_{10}, 0.1)$	ML
	Sure trip between lottery outcomes (1 choice)	
45	A: $(x_5, 1)$ vs. B: $(x_1, 0.1; x_2, 0.1; ...; x_9, 0.1; x_{10}, 0.1)$	–

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Mixture attitudes by probability

	Lottery chosen over x_j (upper)			x_k chosen over lottery (lower)		
	(1)	(2)	(3)	(4)	(5)	(6)
$p = .9$	-6.7^{**}	-5.4^{*}	-5.5^{*}	-0.0	0.6	2.2
	(2.9)	(3.3)	(3.1)	(3.1)	(3.5)	(3.3)
$p = .1$	-17.3^{***}	-18.6^{***}	-18.5^{***}	10.2^{***}	9.6^{**}	8.1^{**}
	(3.1)	(3.4)	(3.6)	(3.6)	(3.9)	(4.1)
Value forgone (£100)		3.3	3.2		-3.3	-11.4
		(3.9)	(2.4)		(7.8)	(7.6)
Problem-triple FE	Yes	Yes	Yes	Yes	Yes	Yes
Respondent FE	No	No	Yes	No	No	Yes
Share at $p = .5$ (%)	27.8	27.8	27.8	12.0	12.0	12.0
Observations	1,113	1,113	1,113	798	798	798
Respondents	79	79	79	79	79	79

Linear probability models, percentage points; reference: $p = .5$. Standard errors clustered by respondent.

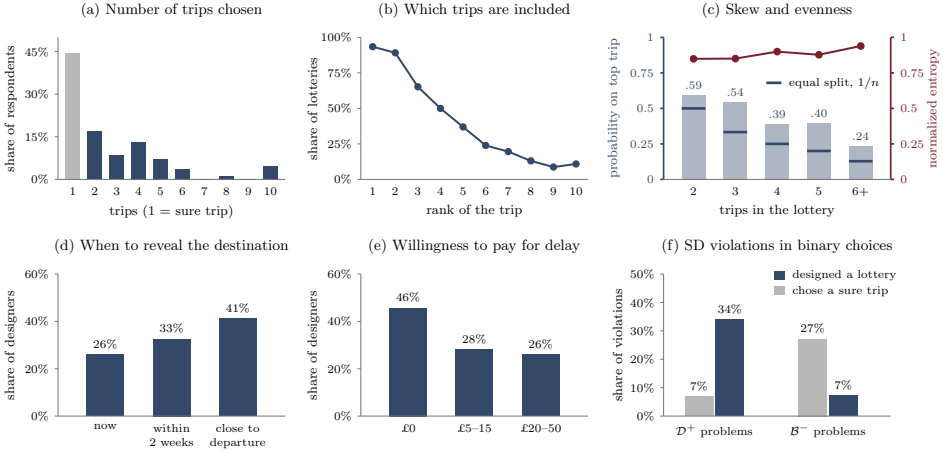
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Noise benchmarks

	Data	10% tremble	Random utility (σ =£50)	Person-specific tremble
Share of respondents by type (%)				
SNB, with a dip	18	51	78	67
		[43, 59]	[72, 84]	[60, 72]
SNB, without a dip	24	19	3	10
		[12, 27]	[0, 6]	[5, 16]
Mixture-loving only	18	7	3	3
		[2, 12]	[0, 6]	[0, 7]
Mixture-averse only	17	5	1	2
		[1, 8]	[0, 2]	[0, 5]
No deviation	17	9	1	11
		[4, 13]	[0, 2]	[7, 14]
Other[a]	6	9	14	8
		[6, 13]	[8, 18]	[5, 11]
Tests				
Type distribution, χ^2		66.6	643.5	196.0
		[1.0, 9.9]	[1.0, 9.4]	[0.6, 8.4]
Correlation of violations, ρ	−0.33	+0.01	+0.23	+0.60
		[-0.17, +0.21]	[+0.05, +0.41]	[+0.48, +0.71]
Out-of-sample gap[b] (pp)	+27.2	+0.1	+5.5	+20.0
		[-6.7, +6.1]	[-17.2, +20.5]	[+14.4, +24.9]

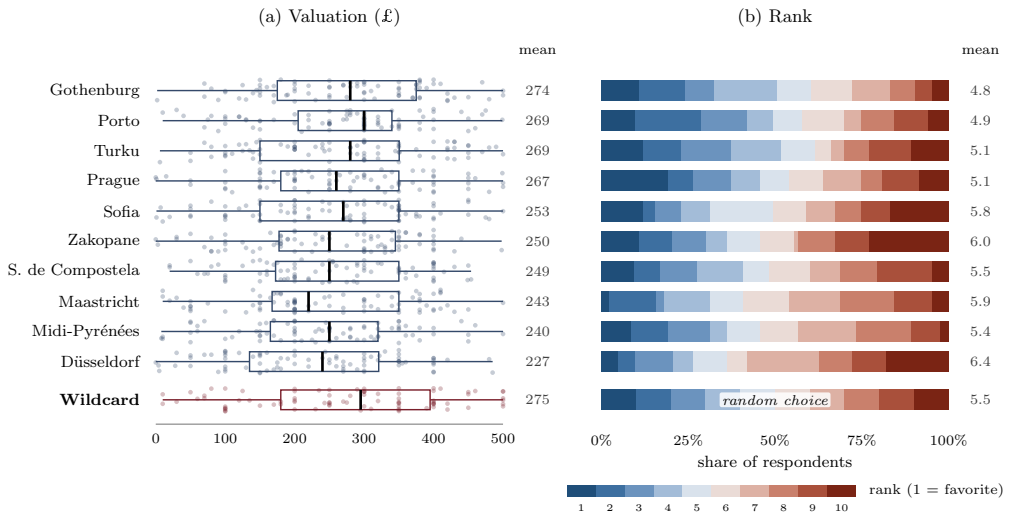
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Lottery building



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No objective winner



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Sample

Participants	83
Recruited through the LSE Behavioural Research Lab	87%
Male	59%
Age, mean (range)	27.1 (18–50)
Trips abroad in the last 12 months, mean	4.0
Of the 10 countries, visited before, mean	4.9
Days to the chosen travel date, mean	230
Completed after the Italian lockdown	51%
Survey time, mean	44 minutes

MMA without WNB: better bundles cost more

Proposition (Informal)

$\succsim$ displays no WNB if and only if, whenever prices $p \gg 0$ support the indifference sets of x and y,

$$x \succ y \quad \Longrightarrow \quad p \cdot x \;\geq\; p \cdot y.$$

Under one-sided attitudes, the better bundle costs more at supporting prices.
MMA without WNB keeps this. WNB is exactly where it fails.